Jinay Patel

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Education

University of Waterloo
Bachelor of Computer Engineering *Expected Graduation: Jun 2031*

Skills

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years)
Technologies: PyTorch, Langchain, Mastra, CMake, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity

Experience

Lead Software Engineer October 2023 — June 2026
Seaquam Robotics *Delta, BC*

- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by +50%
- Created a custom React and Tauri CMS interface, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server response speed by +150%

Embedded Controls Software Engineer October 2021 — June 2026
Seaquam Robotics *Delta, BC*

- Implemented C++ motion algorithms, securing a 100% win rate during autonomous periods across 15+ matches
- Designed an Interpolated Look Up Table using a Monotonic Cubic Spline for tuning PID constants in real-time which resulted in a 200% increase in accuracy across all movements with error tolerances being <1 inch

Software Engineering Tutor September 2025 — March 2026
Code Ninjas *Surrey, BC*

- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving our local branch search visibility

Freelance Systems Administrator / IT Consultant
Convergence Media *Remote*

- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a Tailscale tunnel to expose a public link and allow anyone to access the Nextcloud server for video calls or file sharing

Projects

Posturai | Hackathon Project Devpost

- A Posture correction app that detects bad posture and warn a user about it, built using Python and Streamlit
- Developed a Machine Learning model using PyTorch to classify good and bad posture using a dataset of only 150 images

AI Complaint Categorization System | Personal Project GitHub

- Analyzes multimodal complaints data and categorizes them using of OCR Space, Gemini, and Hugging Face
- Developed a RAG pipeline that stores the data using a vector database and then retrieves it using Langchain

Penetration Testing Agent | Personal Project GitHub

- An agent to perform penetration tests built using Typescript, Bun JS, Mastra, and Puppeteer for scraping
- The agent can scrape tool information, install tools, and run them in the terminal to complete penetration tests

Awards

Excellence Award @ Vex Robotics World Championship, RECF & Vex Robotics April 2026
Awarded to the top team out of 20,000+ teams worldwide for excellent performance

Excellence Award @ Provincial Championship, Vex Robotics February 2026
Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks July 2025
Awarded for developing Posturai, a real-time posture correction software

Euclid Math Contest Distinction, Waterloo CEMC May 2026
Top 25% in Canada for the Euclid math contest